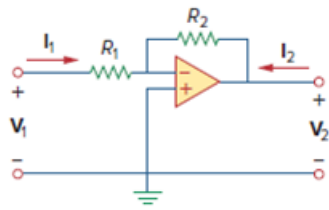


Problem

Find the z parameters of the op amp circuit in Fig. 19.38. Show that the circuit has no y parameters.

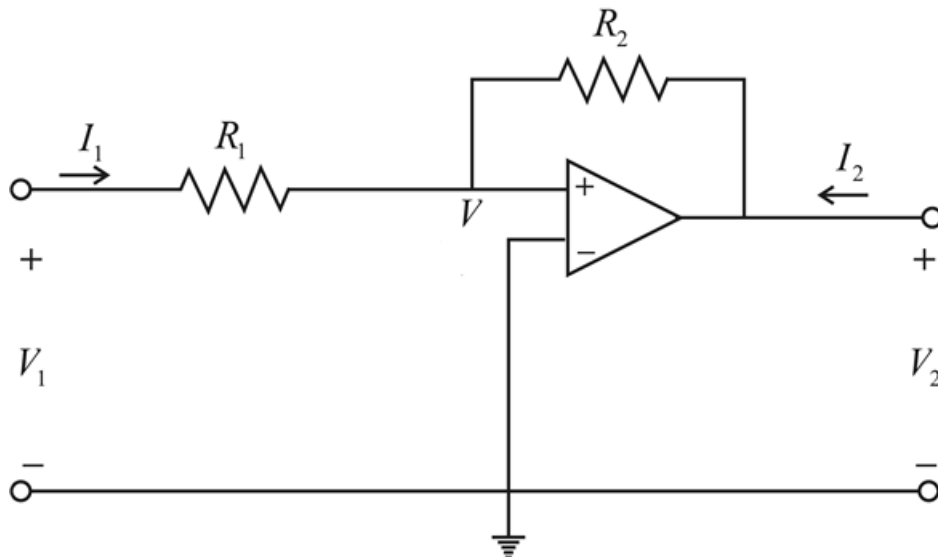
Figure 19.38



Step-by-step solution

Step 1 of 4

Given op amp circuit



Step 2 of 4

Applying nodal analysis at node V

$$\frac{V - V_1}{R_1} + \frac{V - V_2}{R_2} = 0$$

As we know that

$$V^- = V^+$$

$$V = 0$$

Then

$$\frac{-V_1}{R_1} + \frac{-V_2}{R_2} = 0$$

$$\frac{V_2}{R_2} + \frac{V_1}{R_1} = 0$$

$$V_2 = -\frac{R_2}{R_1} V_1 \text{ --- (1)}$$

Thus the input current is

$$I_1 = \frac{V_1}{R_1}$$

$$V_1 = R_1 I_1 \text{ --- (2)}$$

Step 3 of 4

By substituting the value of V_1 from (2) in equation (1) then we obtain

$$V_2 = -R_2 I_1 \text{ --- (3)}$$

we know the basic form of impedance parameters are

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

By comparing the basic equation by (3) and (4) equations are

$$Z_{11} = R_1$$

$$Z_{12} = 0$$

$$Z_{21} = -R_2$$

$$Z_{22} = 0$$

Therefore in matrix form

$$[Z] = \begin{bmatrix} R_1 & 0 \\ -R_2 & 0 \end{bmatrix}$$

Step 4 of 4

The determinant of the $[Z]$ matrix is

$$\Delta_z = Z_{11}Z_{22} - Z_{21}Z_{12}$$

$$\Delta_z = 0$$

Since $\Delta_z = 0$ the $[Z]$ matrix has no inverse therefore $[Z]^{-1}$ does not exist

As we know that

$$[Y] = [Z]^{-1}$$

Hence $[Y]$ matrix doesn't exist

Thus the circuit is not reciprocal because of the active elements